

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

09

Q1

ALGEBRA

Julissa needs at least 100 hours of flight time to get her private pilot certification. If Julissa already has 86 hours of flight time, what is the minimum number of additional hours of flight time Julissa needs to get her private pilot certification?

- A. 14
- B. 76
- C. 86
- D. 186

What is the equation of the line that passes through the point $0, 5$ and is parallel to the graph of $y = 7x + 4$ in the xy -plane?

- A. $y = 5x$
- B. $y = 7x + 5$
- C. $y = 7x$
- D. $y = 5x + 7$

Q3

GRID-IN

ALGEBRA

$$4x + 5 = 165$$

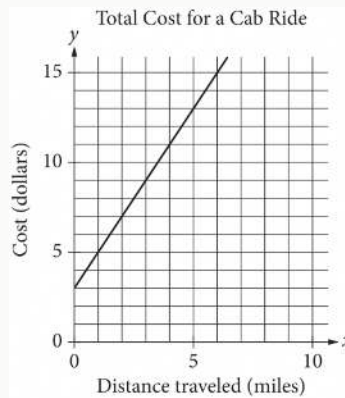
What is the solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

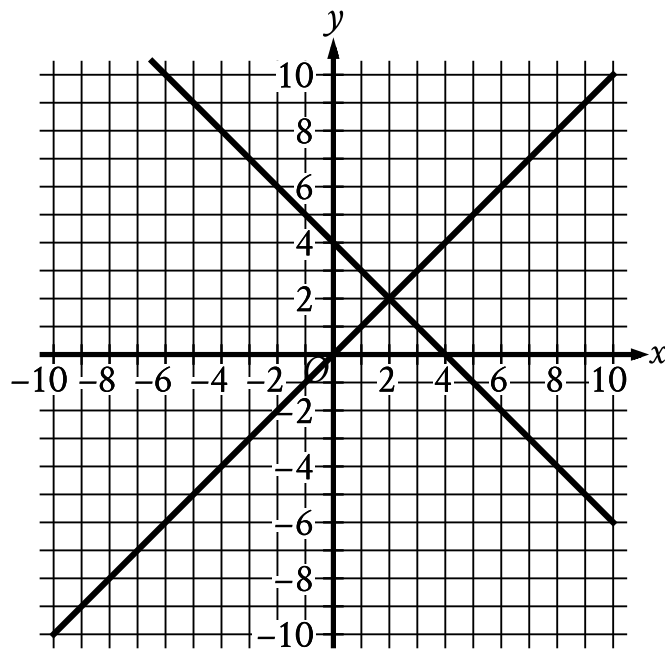
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The line graphed in the xy -plane below models the total cost, in dollars, for a cab ride, y , in a certain city during nonpeak hours based on the number of miles traveled, x .



According to the graph, what is the cost for each additional mile traveled, in dollars, of a cab ride?

- A. \$2.00
- B. \$2.60
- C. \$3.00
- D. \$5.00



- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (0 comma 4)
 - (2 comma 2)
 - (4 comma 0)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (0 comma 0)
 - (2 comma 2)
 - (4 comma 4)

The graph of a system of two linear equations is shown. What is the solution x, y to the system?

A. 0, 4

B. 2,2

C. 4,0

D. 4,4

During spring migration, a dragonfly traveled a minimum of 1,510 miles and a maximum of 4,130 miles between stopover locations. Which inequality represents this situation, where d is a possible distance, in miles, this dragonfly traveled between stopover locations during spring migration?

- A. $d \leq 1,510$
- B. $1,510 \leq d \leq 4,130$
- C. $d \geq 4,130$
- D. $4,130 \leq d \leq 5,640$

Naomi bought both rabbit snails and nerite snails for a total of \$ 52. Each rabbit snail costs \$ 8 and each nerite snail costs \$ 6. If Naomi bought 2 nerite snails, how many rabbit snails did she buy?

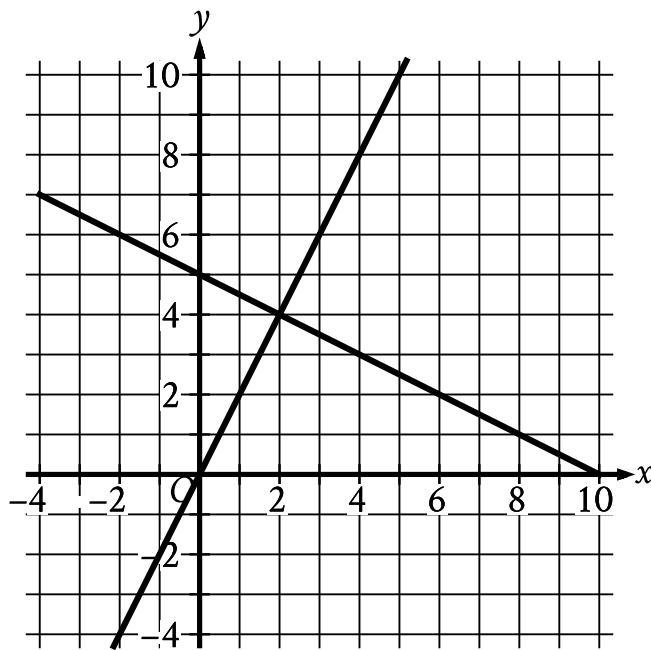
- A. 5
- B. 12
- C. 14
- D. 50

On the first day of a semester, a film club has 90 members. Each day after the first day of the semester, 10 new members join the film club. If no members leave the film club, how many total members will the film club have 4 days after the first day of the semester?

- A. 400
- B. 130
- C. 94
- D. 90

Gabriella deposits \$ 35 in a savings account at the end of each week. At the beginning of the 1st week of a year there was \$ 600 in that savings account. How much money, in dollars, will be in the account at the end of the 4th week of that year?

- A. 460
- B. 635
- C. 639
- D. 740



- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (0 comma 5)
 - (2 comma 4)
 - (10 comma 0)
- For the second line in the system:
 - The line slants sharply up from left to right.
 - The line passes through the following points:
 - (0 comma 0)
 - (2 comma 4)
 - (5 comma 10)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. (0, 5)

B. $(2,4)$

C. $(5,10)$

D. $(10,0)$